Induction validation - Cell Culture

Introduction

Get started by giving your protocol a name and editing this introduction.

Materials

- 12 well plates
- Dox
- TMP
- Media with puro (2 ug/mL)
- 15 mL falcon tubes
- 1.5 mL eppendorf tubes

Procedure

Conditions

1. Biological triplicates of the following conditons

   Total = 30

Table1

	A	B	C	D	E
1		Dox (nM)	TMP(uM)	total media volume needed	total cells needed
2	CRISPRi				
3	GFI1B w/drugs	600	5		
4	GFI1B w/o drugs	0	0		
5	CD46 w/drugs	600	5		
6	CD46 w/o drugs	0	0		
7	NTC w/o drugs	0	0		
8	CRISPRa				
9	GFI1B w/drugs	600	0		
10	GFI1B w/o drugs	0	0		
11	CD46 w/drugs	600	0		
12	CD46 w/o drugs	0	0		
13	NTC w/o drugs	0	0		

Day 0

2. Warm media (RPMI with puro)

3. Count cells

   CRISPRi GFI1B =

   CRISPRi CD46 =

   CRISPRi NTC =

   CRISPRa GFI1B =

   CRISPRa CD46 =

   CRISPRa NTC =

4. Pellet 320,000 cells per condition

   CRISPRi GFi1B t =

   CRISPRi GFi1B ut =

   CRISPRi CD46 t =

   CRISPRi CD46 ut =

   CRISPRi NTC ut =

   CRISPRa GFi1B t =

   CRISPRa GFi1B ut =

   CRISPRa CD46 t =

   CRISPRa CD46 ut =

   CRISPRa NTC ut=

5. Make working diltuions of Dox and TMP

   Add 1 uL of Dox stock (50 mg/mL) to 999 uL media/puro
   Ad 10 uL of TMP stock (100 mM) to 990 uL media/puro

6. Make media + drug mixes

   Add 15 mL of media/puro with 92.4 uL of WD Dox and 75 uL WD TMP

   Add 15 ml of media/puro with 92.4 uL of WD Dox

7. Resuspend with 6.4 mL of respective media

   CRISPRi treated conditions get Dox/TMP/Media/Puro mix

   CRISPRa treated conditions get Dox/Media/Puro mix

   CRISPi/a untreated condtions get normal media/puro mix

8. Plate cells (2 mL per well) and incubate

Day 2

9. Transfer cells to 1.5 mL Eppendorf tubes and spin down at 400 xg for 5 minutes.

10. Make working diltuions of Dox and TMP

    Add 1 uL of Dox stock (50 mg/mL) to 999 uL media/puro

Ad 10 uL of TMP stock (100 mM) to 990 uL media/puro

11. Make condition-specific media for treated conditions

    Add 15 mL of media/puro with 92.4 uL of WD Dox and 75 uL WD TMP

    Add 15 ml of media/puro with 92.4 uL of WD Dox

12. Remove supernatant and resuspend with 1 mL of condition-specific media

13. Take half (0.5 mL) and plate in new well plate

14. Add 1 more mL of condition-specific media

Day 5

15. Transfer cells to 15 mL falcon tubes and spin down at 400 xg for 5 minutes.

16. Wash: Resuspend in 200 uL PBS, spin down at 300 xg for 5 min at 22C and remove supernatant

17. Resuspend cells in 350 uL RLT buffer

18. Transfer to Eppendorf tubes and freeze -80C (add to dry ice first)

--> qPCR readout